Lumbar Laminectomy

ORG: S-830 (ISC)
Link to Codes

MCG Health
Inpatient & Surgical Care
29th Edition

Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Procedure

- Procedure is indicated for **1 or more** of the following(1)(2):
  - Spinal cord compression (myelopathy), as indicated by **ALL** of the following(3)(4)(5):
    - Progressive or severe neurologic deficits consistent with spinal cord compression (eg, bladder or bowel incontinence)
    - Imaging findings of lumbar cord compression that correlate with clinical findings
  - Cauda equina syndrome, as indicated by **ALL** of the following(5)(6):
    - Progressive or severe neurologic deficits consistent with cauda equina or spinal cord compression, including **1 or more** of the following:
      - Bowel dysfunction
      - Bladder dysfunction (eg, incontinence or urinary retention)
      - Saddle anesthesia
      - Bilateral lower extremity neurologic abnormalities
      - New decrease in rectal tone or sacral reflexes
    - MRI or other neuroimaging finding that correlates with clinical signs and symptoms
  - Lumbar spinal stenosis, as indicated by **1 or more** of the following(1)(7)(8)(17):
    - Rapidly progressive or very severe symptoms of neurogenic claudication with imaging findings of lumbar spinal stenosis that correlate with clinical findings
    - Leg or buttock neurogenic claudication symptoms and **ALL** of the following:
      - Symptoms that are persistent and disabling
      - Imaging findings of lumbar spinal stenosis that correlate with clinical findings
      - Failure of 3 months of nonoperative therapy(18)
  - Lumbar spondylolisthesis, as indicated by **1 or more** of the following(2)(9)(10)(19):
    - Rapidly progressive or severe neurologic deficits (eg, bowel or bladder dysfunction)
    - Symptoms requiring treatment, as indicated by **ALL** of the following:
      - Patient has persistent disabling symptoms, including **1 or more** of the following:
        - Low back pain
        - Neurogenic claudication

            - Radicular pain
        - Treatment is indicated by **ALL** of the following:
            - Listhesis demonstrated on imaging
            - Symptoms that correlate with findings on MRI or other imaging
            - Failure of 3 months of nonoperative therapy
    - Lumbar disk disease and **ALL** of the following(1):
        - Patient has unremitting radicular pain or progressive weakness secondary to nerve root compression.
        - Imaging findings of lumbar disk disease that correlate with clinical findings
        - Failure of 6 weeks of nonoperative therapy that includes **1 or more** of the following:
            - Medication (eg, NSAIDs, analgesics)
            - Physical therapy
            - Epidural or oral corticosteroid
- Dorsal rhizotomy for spasticity (eg, cerebral palsy)(12)
- Signs or symptoms of lumbar disease (eg, pain, motor weakness, bowel or bladder incontinence) secondary to tumor or neoplasm(13)
- Signs or symptoms of lumbar disease (eg, pain, motor weakness, bowel or bladder incontinence) secondary to infectious process (eg, epidural abscess)(14)(20)
- Signs or symptoms of lumbar disease (eg, pain, motor weakness, bowel or bladder incontinence) secondary to acute trauma(15)(16)

Alternatives to Procedure

- Alternatives include(1)(2)(5)(8)(9)(10)(16):
    - Nonoperative measures, including(5)(17)(23):
        - Anti-inflammatory medication
        - Analgesics
        - Muscle relaxants
        - Lumbar bracing
        - Physical therapy
        - Epidural or oral corticosteroid(24)(25)
    - Laminectomy with fusion. See Lumbar Fusion ISC guideline.
    - Diskectomy without laminectomy. See Lumbar Diskectomy, Foraminotomy, or Laminotomy ISC guideline.
    - Percutaneous rhizotomy. See Rhizotomy, Percutaneous ISC.
    - Hemilaminectomy
    - Multilevel unilateral or bilateral laminotomy (microdecompression)(26)
    - Radiation therapy, chemotherapy, and glucocorticoids (eg, for neoplastic epidural compression)
    - Percutaneous drainage and antibiotics for epidural abscess(27)(28)

Operative Status Criteria

Goal Length of Stay: Ambulatory

Note: The definition of an ambulatory procedure depends on payer-provider contractual agreement or regulatory language (eg, CMS' Two-Midnight Rule). An ambulatory procedure may include one postoperative overnight stay in a facility; therefore, MCG's ambulatory Goal Length of Stay (GLOS) attainment calculation reports the sum of same-day and next-day postoperative discharges. Depending on various patient and procedural factors, some patients undergoing a procedure with an ambulatory GLOS require inpatient care (eg, medical necessity for hospital-based care across 2 or more postoperative midnights). Some of these factors are described in the Extended Stay section of this guideline.

- Ambulatory
- Inpatient (eg, medical necessity for hospital-based care across 2 or more postoperative midnights)
- Inpatient (Medicare patient, and specific procedure is on CMS Inpatient Only List)

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
1	• Social Determinants of Health Assessment	• **Hemodynamic stability** • **No new neurologic**	• **Ambulatory or acceptable**	• IV fluids • IV medications • **Oral hydration[B]**	• Avoid lifting, bending, and prolonged sitting	• Prophylactic antibiotics[C] • Analgesics

	• OR to recovery room to discharge[A] • Discharge planning	**deficits** • **Voiding ability at baseline** • **No evidence of infection** • **Pain absent or managed** • **Discharge plans and education understood**	**for next level of care**	• **Oral medications or regimen acceptable for next level of care** • **Oral diet or acceptable for next level of care**		

(1)(30)(31)(32)(33)

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: Ambulatory

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Inpatient stay (eg, need for hospital-based care beyond postoperative day 1) may be needed for(1)(17)(34):
  - Failure to meet discharge criteria (recovery milestones in Optimal Recovery Course)
    - Expect brief stay extension.
  - Severe preoperative deficit or injury
    - Expect brief stay extension.
  - Infection
    - Patient with infectious basis for surgery may require longer monitoring on parenteral antibiotics and confirmation of culture results.
    - Expect brief stay extension.
  - Fracture or neoplasm(35)
    - Expect brief stay extension.
  - Active comorbid illness (eg, cardiovascular or pulmonary disease, diabetes)(31)
    - Expect brief stay extension.
  - Dural tear or intradural procedure(31)(33)(36)
    - Expect brief stay extension.
  - Intraspinal hemorrhage
    - Expect brief stay extension.
  - Extraspinal hemorrhage
    - Expect brief stay extension.
  - Nerve root injury
    - Expect brief stay extension.
  - Pediatric patient[D]
    - Expect brief stay extension.
  - Postoperative altered mental status (eg, delirium)
    - Anticipate evaluation for etiology (eg, electrolyte abnormalities, pain, medications) with specific management as appropriate.
    - Expect brief stay extension.

See Common Complications and Conditions ISC for further information.

Discharge

Discharge Planning

- Discharge planning includes[E]:
  - Assessment of needs and planning for care, including(38)(39):
    - Develop and modify treatment plan (involving multiple providers) as needed.
    - Evaluate and address preadmission functioning as needed.

    - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment SR for further information.
    - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool SR for further information.(37)
    - Evaluate and address patient or caregiver preferences as indicated.
    - Identify skilled services needed at next level of care, with specific attention to(40):
      - Neurologic status assessment(41)
      - Pain management
      - Wound or dressing management
  - Early identification of anticipated discharge destination; options include(39)(42):
    - Home; considerations include:
      - Home safety assessment. See Home Safety Assessment SR for further information.
      - Patient safe to go home; examples include(43)(44)(45):
        - Medical status stable for patient's condition
        - Functional care can safely be provided with available resources.
        - Mental status stable for patient's condition
        - Medication availability confirmed and reconciliation complete
        - Patient/caregiver education completed with written discharge instructions provided
        - Community resources identified and referrals made, as needed
        - Home care arranged, if indicated
        - Necessary medical equipment delivery arranged or available in home, if indicated
        - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
      - Access to follow-up care
      - Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment SR for further information.
      - Caregiver need, ability, and availability
    - Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool SR for further information.
  - Transitions of care plan complete, including(39)(42)(46):
    - Patient and caregiver education complete.
      - See Teach Back Tool SR for further information.
      - See Lumbar Laminectomy: Patient Education for Clinicians SR for further information.
    - Medication reconciliation completion includes(47)(48):
      - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
      - Assess each medication for correlation to disease state or medical condition.
      - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
      - Provide reconciled medication list to all treating providers.
      - Confirm that patient or caregiver can acquire medication.
      - Educate patient and caregiver.
        - Provide complete medication list to patient and caregiver.
        - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
        - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(49)
      - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
      - See Medication Reconciliation Tool SR for further information.
    - Plan communicated to patient, caregiver, and all members of care team, including(50):
      - Inpatient care and service providers
      - Primary care provider
      - All post-discharge care and service providers
    - Appointments planned or scheduled, which may include(41):
      - Primary care provider
      - Neurosurgeon
      - Orthopedic surgeon
      - Rehabilitation therapy services(51)
      - Specialists for management of comorbidities as needed
      - Other
    - Outpatient testing and procedure plans made, which may include:
      - Other

- Referrals made for assistance or support, which may include:
  - Financial, for follow-up care, medication, and transportation(52)
  - Tobacco use treatment(53)
  - Vocational rehabilitation(41)
  - Other
- Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include:
  - Ambulation devices (eg, cane, crutches, walker)(54)
  - Wound care equipment and supplies(55)
  - Other

Discharge Destination

- Post-hospital levels of admission may include:
  - Home.
  - Home healthcare. See Home Care Indications for Admission Section HC in Lumbar Spine Surgery guideline in Home Care.
  - Recovery facility care. See Recovery Facility Care Indications for Admission Section RFC in Lumbar Spine Surgery guideline in Recovery Facility Care.

Evidence Summary

Criteria

The evidence for the clinical indications found in this guideline includes 17 published peer reviewed articles and 2 book sections.

For spinal cord compression (myelopathy), an orthopedic surgery textbook states that operative management with laminectomy is the treatment of choice for progressive or severe neurologic deficits (eg, bladder or bowel incontinence) in patients with correlating imaging findings.(1) **(EG 2)** A narrative review states that spinal cord compression due to an ossified ligamentum flavum should be treated with laminectomy if symptoms are progressive.(3) **(EG 2)** A narrative review states that myelopathy associated with ossification of the posterior longitudinal ligament should be treated with laminectomy.(4) **(EG 2)**

For cauda equina syndrome, an orthopedic surgery textbook and a narrative review support emergent lumbar laminectomy for patients with neuroimaging findings and progressive or severe neurologic deficits (eg, bladder or bowel dysfunction, saddle anesthesia, bilateral lower extremity neurologic abnormalities).(1)(5) **(EG 2)** A narrative review states that patients with MRI findings and symptoms correlating with cauda equina syndrome should have a surgical decompression procedure within 48 hours of the onset of symptoms.(6) **(EG 2)**

For lumbar spinal stenosis, an orthopedic surgery textbook and narrative reviews state that laminectomy is the procedure of choice for patients with symptoms of leg or buttock neurogenic claudication, correlative MRI findings, and failure of nonoperative therapy.(1)(7)(8) **(EG 2)**

For lumbar spondylolisthesis, an orthopedic surgery textbook and a narrative review support laminectomy as an option for surgical decompression.(2)(9) **(EG 2)** Another narrative review states that surgical intervention, such as laminectomy, is indicated when patients have symptoms of rapidly progressive or severe neurologic deficits (eg, bowel or bladder dysfunction) or if symptoms fail to respond to at least 3 months of nonoperative therapy.(10) **(EG 2)**

For lumbar disk disease, an orthopedic surgery textbook and a narrative review support operative management, such as lumbar laminectomy, if patients have unremitting radicular pain secondary to nerve root compression, imaging findings that correlate with those symptoms, and failure of at least 6 weeks of nonoperative therapy (eg, anti-inflammatory medication, physical therapy, epidural corticosteroid).(1)(11) **(EG 2)**

For dorsal rhizotomy for spasticity (eg, cerebral palsy), a technology appraisal supports laminectomy for selective dorsal rhizotomy to improve muscle tone and gross motor function.(12) **(EG 2)**

For signs or symptoms of lumbar disease (eg, pain, motor weakness, bowel or bladder incontinence) secondary to tumor, infectious process, or acute trauma, narrative reviews and a systematic review support surgical management, including laminectomy, as an appropriate part of the treatment regimen.(13)(14)(15)(16) **(EG 2)**

Alternatives

A systematic review and meta-analysis of 6 studies, encompassing 531 patients (mean age 66 years) with spinal stenosis and degenerative lumbar spondylolisthesis randomized to treatment with lumbar decompression alone (73% laminectomy, 14% laminotomy, 13% microendoscopic decompression) or lumbar decompression with fusion, found (over a mean 27-month follow-up period) no significant differences in back pain (mean difference (MD) 0.24, 95% confidence interval (CI) -0.38 to 0.85), leg pain (MD 0.39, 95% CI -0.34 to 1.11), or disability score (MD 0.50, 95% CI -3.31 to 4.31); however, the decompression-only group had fewer complications (odds ratio (OR) 0.57, 95% CI 0.36 to 0.90) but more subsequent postoperative degenerative spondylolisthesis (OR 3.49, 95% CI 1.05

to 11.65).(21) **(EG 1)** A study of 211 patients (mean age 66 years) with lumbar spinal stenosis randomized to 1-level or 2-level decompression alone or decompression with fusion found that spinal restenosis was less frequent after decompression with fusion at the operative level (4% vs 14% for decompression with fusion vs decompression alone, respectively); however, this was more than offset by an increase in the rate of new stenosis at an adjacent level (44% vs 17%).(22) **(EG 1)**

Length of Stay

Analysis of 194 patients (mean age 67 years) undergoing lumbar laminectomy showed a mean length of stay of 1.2 days.(32) **(EG 2)** Analysis of procedure data for a commercially insured population shows 86% of patients undergoing lumbar laminectomy were discharged the day of or the day after surgery.(33) **(EG 3)** Analysis of procedure data for a Medicare-insured population shows 84% of patients undergoing lumbar laminectomy were discharged the day of or the day after surgery.(33) **(EG 3)**

Rationale

Use of this MCG care guideline helps the clinician identify, for a given procedure, which patient-specific factors and clinical conditions are appropriate for that procedure. The evidence-based clinical criteria assist the clinician in the decision to appropriately perform a procedure, evaluating whether the potential benefits of a procedure outweigh the potential risks. For Medicare enrollees, surgical MCG care guidelines also identify which procedures CMS has designated as inpatient only.

Use of these evidence-based clinical criteria to support decision making around the need for a given procedure is of benefit to the patient, as all procedures come with inherent risk that must be balanced by anticipated clinical benefit. Utilizing evidence-based clinical criteria enables a more accurate and patient-specific decision-making process. In addition, the use of evidence-based guidelines can help reduce unwarranted variation in care, such as divergent clinical thresholds to perform a procedure for clinically similar patients that vary across geographic regions, between facilities, and among individual clinicians.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 412.3(56); 42 CFR 419.22(57); 42 CFR 422.101(58)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 1 - Inpatient Hospital Services Covered Under Part A(59); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 6 - Hospital Services Covered Under Part B(60); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(61); CMS IOM Publication 100-08, Medicare Program Integrity Manual, Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment(62)

Medicare Coverage Determinations: Medicare Coverage Database(63)

References

1. Gardocki RJ, Park AL. Degenerative disorders of the thoracic and lumbar spine. In: Azar FM, Beaty JH, editors. Campbell's Operative Orthopaedics. 14th ed. Philadelphia, PA: Elsevier; 2021:1719-1801 e9. [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
2. Williams KD. Spondylolisthesis. In: Azar FM, Beaty JH, editors. Campbell's Operative Orthopaedics. 14th ed. Philadelphia, PA: Elsevier; 2021:1802-1831.e1. [Context Link 1, 2, 3, 4]
3. Daniels AH, McDonald CL, Basques BA, Kuris EO. Ossified Ligamentum Flavum: epidemiology, treatment, and outcomes. Journal of the American Academy of Orthopedic Surgeons 2022;30(12):e842-e851. DOI: 10.5435/JAAOS-D-21-01253. [Context Link 1, 2] View abstract...
4. Le HV, Wick JB, Van BW, Klineberg EO. Ossification of the posterior longitudinal ligament: pathophysiology, diagnosis, and management. Journal of the American Academy of Orthopedic Surgeons 2022;30(17):820-830. DOI: 10.5435/JAAOS-D-22-00049. [Context Link 1, 2] View abstract...
5. Zhang AS, et al. Lumbar disc herniation: diagnosis and management. American Journal of Medicine 2023;136(7):645-651. DOI: 10.1016/j.amjmed.2023.03.024. [Context Link 1, 2, 3, 4, 5] View abstract...
6. Bulloch L, Thompson K, Spector L. Cauda equina syndrome. Orthopedic Clinics of North America 2022;53(2):247-254. DOI: 10.1016/j.ocl.2021.11.010. [Context Link 1, 2] View abstract...
7. Lafian AM, Torralba KD. Lumbar spinal stenosis in older adults. Rheumatic Diseases Clinics of North America 2018;44(3):501-512. DOI: 10.1016/j.rdc.2018.03.008. [Context Link 1, 2] View abstract...
8. Katz JN, Zimmerman ZE, Mass H, Makhni MC. Diagnosis and management of lumbar spinal stenosis: a review. Journal of the American Medical Association 2022;327(17):1688-1699. DOI: 10.1001/jama.2022.5921. [Context Link 1, 2, 3] View abstract...
9. Karsy M, Bisson EF. Surgical versus nonsurgical treatment of lumbar spondylolisthesis. Neurosurgery Clinics of North America 2019;30(3):333-340. DOI: 10.1016/j.nec.2019.02.007. [Context Link 1, 2, 3] View abstract...
10. Bydon M, Alvi MA, Goyal A. Degenerative lumbar spondylolisthesis: definition, natural history, conservative management, and surgical treatment. Neurosurgery Clinics of North America 2019;30(3):299-304. DOI: 10.1016/j.nec.2019.02.003. [Context Link 1, 2, 3] View abstract...
11. Foster NE, et al. Prevention and treatment of low back pain: evidence, challenges, and promising directions. Lancet 2018;391(10137):2368-2383. DOI: 10.1016/S0140-6736(18)30489-6. [Context Link 1] View abstract...

12. Selective Dorsal Rhizotomy for Spasticity in Cerebral Palsy. NICE Interventional Procedure Guidance IPG373 [Internet] National Institute for Health and Care Excellence. 2010 Dec Accessed at: https://www.nice.org.uk/guidance/. [accessed 2024 Oct 01] [Context Link 1, 2]
13. Nieblas-Bedolla E, Zuccato J, Kluger H, Zadeh G, Brastianos PK. Central nervous system metastases. Hematology/Oncology Clinics of North America 2022;36(1):161-188. DOI: 10.1016/j.hoc.2021.08.004. [Context Link 1, 2] View abstract...
14. Narayan V, Mohammed N, Savardekar AR, Patra DP, Nanda A. Tuberculous spondylolisthesis: a reappraisal of the clinicoradiologic spectrum and surgical treatment paradigm. World Neurosurgery 2018;114:361-367. DOI: 10.1016/j.wneu.2018.02.157. [Context Link 1, 2] View abstract...
15. Wallace DJ, Sy C, Peitz G, Grandhi R. Management of non-missile penetrating spinal injury. Neurosurgical Review 2019;42(4):791-798. DOI: 10.1007/s10143-018-01057-1. [Context Link 1, 2] View abstract...
16. Waddell WH, Gupta R, Stephens BF. Thoracolumbar spine trauma. Orthopedic Clinics of North America 2021;52(4):481-489. DOI: 10.1016/j.ocl.2021.05.014. [Context Link 1, 2, 3] View abstract...
17. Darlow M, et al. A pathway for the diagnosis and treatment of lumbar spinal stenosis. Orthopedic Clinics of North America 2022;53(4):523-534. DOI: 10.1016/j.ocl.2022.05.006. [Context Link 1, 2, 3] View abstract...
18. Jarrett MS, Orlando JF, Grimmer-Somers K. The effectiveness of land based exercise compared to decompressive surgery in the management of lumbar spinal-canal stenosis: a systematic review. BMC Musculoskeletal Disorders 2012;13:30. DOI: 10.1186/1471-2474-13-30. [Context Link 1] View abstract...
19. McGowan JE, Kanter AS. Lateral approaches for the surgical treatment of lumbar spondylolisthesis. Neurosurgery Clinics of North America 2019;30(3):313-322. DOI: 10.1016/j.nec.2019.02.005. [Context Link 1] View abstract...
20. Chaker AN, et al. Epidural abscess: a propensity analysis of surgical treatment strategies. Spine 2018;43(24):E1479-E1485. DOI: 10.1097/BRS.0000000000002747. [Context Link 1] View abstract...
21. Abdel-Fattah AR, et al. To fuse or not to fuse: The elderly patient with lumbar stenosis and low-grade spondylolisthesis. Systematic review and meta-analysis of randomised controlled trials. Surgeon 2023;21(1):Online. DOI: 10.1016/j.surge.2022.02.008. [Context Link 1] View abstract...
22. Karlsson T, et al. Decompression alone or decompression with fusion for lumbar spinal stenosis: a randomized clinical trial with two-year MRI follow-up. Bone and Joint Journal 2022;104-B(12):1343-1351. DOI: 10.1302/0301-620X.104B12.BJJ-2022-0340.R1. [Context Link 1] View abstract...
23. Zaina F, Tomkins-Lane C, Carragee E, Negrini S. Surgical versus non-surgical treatment for lumbar spinal stenosis. Cochrane Database of Systematic Reviews 2016, Issue 1. Art. No.: CD010264. DOI: 10.1002/14651858.CD010264.pub2. [Context Link 1] View abstract...
24. Chou R, et al. Systemic corticosteroids for radicular and non-radicular low back pain. Cochrane Database of Systematic Reviews 2022, Issue 10. Art. No.: CD012450. DOI: 10.1002/14651858.CD012450.pub2. [Context Link 1] View abstract...
25. Smith CC, McCormick ZL, Mattie R, MacVicar J, Duszynski B, Stojanovic MP. The effectiveness of lumbar transforaminal injection of steroid for the treatment of radicular pain: a comprehensive review of the published data. Pain Medicine 2020;21(3):472-487. DOI: 10.1093/pm/pnz160. [Context Link 1] View abstract...
26. Lee CY, et al. Minimally invasive surgical treatment of extensive spinal epidural abscess with unilateral laminotomy for bilateral decompression using an ultrasonic bone curette: a technique note. World Neurosurgery 2022;168:111-119. DOI: 10.1016/j.wneu.2022.09.089. [Context Link 1] View abstract...
27. Gardner WT, Rehman H, Frost A. Spinal epidural abscesses - The role for non-operative management: A systematic review. Surgeon 2021;19(4):226-237. DOI: 10.1016/j.surge.2020.06.011. [Context Link 1] View abstract...
28. Saleh ES, Vasileff CC, Omari AM, Khalil JG. The diagnosis and management of pediatric spine infections. Cureus 2021;13(7):Online. DOI: 10.7759/cureus.16748. [Context Link 1] View abstract...
29. Shaffer WO, Baisden JL, Fernand R, Matz PG, North American Spine Society. An evidence-based clinical guideline for antibiotic prophylaxis in spine surgery. Spine Journal 2013;13(10):1387-1392. DOI: 10.1016/j.spinee.2013.06.030. [Context Link 1] View abstract...
30. Mooney J, et al. Outpatient versus inpatient lumbar decompression surgery: a matched noninferiority study investigating clinical and patient-reported outcomes. Journal of Neurosurgery: Spine 2022;37(4):485-497. DOI: 10.3171/2022.3.SPINE211558. [Context Link 1] View abstract...
31. Dodo Y, et al. Risk factors for ambulatory surgery conversion to extended stay among patients undergoing one-level or two-level posterior lumbar decompression. Spine 2023;48(11):748-757. DOI: 10.1097/BRS.0000000000004621. [Context Link 1, 2, 3] View abstract...
32. Finneran MM, Johnson R, Guglielmi G, Nardone E. Minimally invasive bilateral decompressive lumbar laminectomy with unilateral approach among obese versus nonobese patients. World Neurosurgery 2023;179:e81-e89. DOI: 10.1016/j.wneu.2023.08.015. [Context Link 1, 2] View abstract...
33. Proprietary health insurance data sources (2022-2023); and Medicare 5% Standard Analytical File (2021-2022). [Context Link 1, 2, 3, 4, 5]
34. Premier PINC AI™ Healthcare Database (PHD), 01/01/2022-12/31/2023. Premier, Inc. [Context Link 1]
35. Bhimani AD, et al. Preoperative steroids do not improve outcomes for intramedullary spinal tumors: a NSQIP analysis of 30-day reoperation and readmission rates. Journal of Spine Surgery (Hong Kong) 2018;4(1):9-16. DOI: 10.21037/jss.2018.03.18. [Context Link 1] View abstract...
36. Hassanzadeh H, Bell J, Bhatia M, Puvanesarajah V. Incidental durotomy in lumbar spine surgery; risk factors, complications, and perioperative management. Journal of the American Academy of Orthopedic Surgeons 2021;29(6):e279-e286. DOI: 10.5435/JAAOS-D-20-00210. [Context Link 1] View abstract...
37. Hudson T. The role of social determinates of health in discharge practices. Nursing Clinics of North America 2021;56(3):369-378. DOI: 10.1016/j.cnur.2021.04.004. [Context Link 1, 2] View abstract...
38. Chovanec KA, Arsene C, Beck A, Zachrich K, Liedel B, Wolff-Elliott J. Association of discharge disposition with outcomes. Population Health Management 2021;24(1):116-121. DOI: 10.1089/pop.2019.0176. [Context Link 1] View abstract...

39. Centers for Medicare and Medicaid Services. "Condition of participation: Discharge planning." 42 CFR Pt. 482.43 Washington, DC 2022 Oct [accessed 2024 Oct 08] Accessed at: https://www.ecfr.gov/. [Context Link 1, 2, 3]
40. Snyder JS, Sump CA. Intervertebral Disk Disease. In: Snyder JS, Sump CA, editors. Swearingen's All-in-One Nursing Care Planning Resource. 6th ed. Elsevier; 2024:280-290. [Context Link 1]
41. Filson R. Musculoskeletal trauma and orthopedic surgery. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:1638-1672. [Context Link 1, 2, 3]
42. Targeted care planning and care transitions. In: Perez R, editor. CMSA's Integrated Case Management. 2nd ed. Springer Publishing Company LLC; 2023:178-189. [Context Link 1, 2]
43. Elements of Excellence in Transitions of Care (TOC). TOC Checklist [Internet] National Transitions of Care Coalition. Accessed at: https://www.ntocc.org. [accessed 2024 Sep 24] [Context Link 1]
44. Medical-surgical nursing. In: Hinkle JL, Cheever KH, Overbaugh KJ, editors. Brunner & Suddarth's Textbook of Medical-Surgical Nursing. 15th ed. Wolters Kluwer; 2022:33-55. [Context Link 1]
45. Transition of care. In: Gillingham DC, editor. Foundations of Case Management. Blue Bayou Press; 2021:137-144. [Context Link 1]
46. Becker C, et al. Interventions to improve communication at hospital discharge and rates of readmission: a systematic review and meta-analysis. JAMA Network Open 2021;4(8):e2119346. DOI: 10.1001/jamanetworkopen.2021.19346. [Context Link 1] View abstract...
47. The nursing process in drug therapy and patient safety. In: Tucker RG, editor. Karch's Focus on Nursing Pharmacology. 9th ed. Wolters Kluwer; 2023:47-57. [Context Link 1]
48. Adverse drug reactions and medication errors. In: Burchum JR, Rosenthal LD, editors. Lehne's Pharmacology for Nursing Care. 12th ed. Elsevier; 2025:64-75. [Context Link 1]
49. Safe medication preparation. In: Perry AG, Potter PA, Ostendorf WR, Laplante N, editors. Clinical Nursing Skills and Techniques. 11th ed. Elsevier; 2025:586-607. [Context Link 1]
50. The case manager's roles, functions, and activities. In: Gillingham DC, editor. Foundations of Case Management. Blue Bayou Press; 2021:9-14. [Context Link 1]
51. The intervention. In: Dutton M, editor. Dutton's Orthopaedic Examination, Evaluation, and Intervention. 6th ed. McGraw-Hill Education; 2023:343-368. [Context Link 1]
52. Bragg-Underwood T. Family health. In: Nies MA, editor. Community/Public Health Nursing. 8th ed. Elsevier; 2024:404-427. [Context Link 1]
53. Harding MM. Substance use disorders in acute care. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:162-178. [Context Link 1]
54. Activity. In: Lynn P, editor. Taylor's Clinical Nursing Skills. 6th ed. Wolters Kluwer; 2023:541-632. [Context Link 1]
55. Management of patients with dermatologic disorders. In: Hinkle JL, Cheever KH, Overbaugh KJ, editors. Brunner & Suddarth's Textbook of Medical-Surgical Nursing. 15th ed. Wolters Kluwer; 2022:1815-1864. [Context Link 1]
56. Centers for Medicare and Medicaid Services. "Admissions." 42 CFR 412.3 Washington, DC 2023 Jul [accessed 2024 Oct 08] Accessed at: https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-412. [Context Link 1]
57. Centers for Medicare and Medicaid Services. "Hospital services excluded from payment under the hospital outpatient prospective payment system." 42 CFR 419.22 Washington, DC 2023 Jul [accessed 2024 Oct 08] Accessed at: https://www.ecfr.gov/. [Context Link 1]
58. Centers for Medicare and Medicaid Services. "Requirements relating to basic benefits." 42 CFR 422.101 Washington, DC 2024 Sep 30 [accessed 2024 Oct 08] Accessed at: https://www.ecfr.gov/. [Context Link 1]
59. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 1 - Inpatient Hospital Services Covered Under Part A Rev. 10892 [Internet] Centers for Medicare and Medicaid Services. 2021 Aug Accessed at: https://www.cms.gov/manuals/. [accessed 2024 Jan 18] [Context Link 1]
60. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 6 - Hospital Services Covered Under Part B Rev. 12425 [Internet] Centers for Medicare and Medicaid Services. 2023 Dec Accessed at: http://www.cms.gov/manuals/. [accessed 2024 Jun 05] [Context Link 1]
61. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 15 - Covered Medical and Other Health Services Rev. 12425 [Internet] Centers for Medicare & Medicaid Services. 2023 Dec 21 Accessed at: https://www.cms.gov/manuals/. [accessed 2024 Feb 01] [Context Link 1]
62. Centers for Medicare and Medicaid Services. Medicare Program Integrity Manual. Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment Rev. 10365 [Internet] Centers for Medicare and Medicaid Services. 2020 Oct 02 Accessed at: https://www.cms.gov/medicare/regulations-guidance/manuals/internet-only-manuals-ioms. [accessed 2024 Sep 26] [Context Link 1]
63. Medicare Coverage Database. [Internet] Centers for Medicare and Medicaid Services. Accessed at: https://www.cms.gov/medicare-coverage-database/search.aspx? Updated 2024 [accessed 2024 Sep 26] [Context Link 1]

Footnotes

[A] For ambulatory operative status criteria patients, see Ambulatory Surgery Discharge and Complications: Common Complications and Conditions ISC as needed. [A in Context Link 1]

[B] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [B in Context Link 1]

[C] Prophylactic antibiotics are suggested to decrease the rate of spinal infections following uninstrumented lumbar spinal surgery.(29) [C in Context Link 1]

[D] The most common pediatric procedure (accounting for nearly 60% of pediatric lumbar laminectomies) is laminectomy with release of tethered spinal cord. This procedure is rarely performed on an outpatient basis.(33) [D in Context Link 1]

[E] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(37) [E in Context Link 1]

Definitions

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
  - Hemodynamic abnormalities at baseline or acceptable for next level of care
  - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
    - Tachycardia absent
    - Hypotension absent
    - No evidence of inadequate perfusion (eg, no myocardial ischemia)
    - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

References

1. Puskarich MA, Jones AE. Shock. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:34-41.e1.
2. Lewis J, Patel B. Shock. In: Gershel JC, Rauch DA, editors. Caring for the Hospitalized Child: A Handbook of Inpatient Pediatrics. 3rd ed. Elk Grove Village, IL: American Academy of Pediatrics; 2023:107-113.
3. Ingbar DH, Thiele H. Cardiogenic shock and pulmonary edema. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2250-2257.
4. Brant EB, Seymour CW, Angus DC. Sepsis and septic shock. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2241-2249.
5. Singer M, et al. The Third International Consensus definitions for sepsis and septic shock (Sepsis-3). Journal of the American Medical Association 2016;315(8):801-810. DOI: 10.1001/jama.2016.0287.

Hypotension absent

- Hypotension absent[A], as indicated by **1 or more** of the following(1)(2)(3)(4)(5):
  - SBP greater than or equal to 90 mm Hg in adult or child age 10 years or older
  - Mean arterial pressure[B] greater than or equal to 70 mm Hg in adult or child age 10 years or older MAP Calculator
  - Mean arterial pressure[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure) MAP Calculator
  - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
  - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

References

1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Cooney KA, editors. Goldman-Cecil Medicine. 27th ed. Elsevier; 2024:32-35.
2. Angus DC. Approach to the patient with shock. In: Goldman L, Cooney KA, editors. Goldman-Cecil Medicine. 27th ed. Elsevier; 2024:658-665.
3. Massaro AF. Approach to the patient with shock. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2235-2241.
4. Horeczko T. Pediatric cardiac disorders. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:2109-2131.e1.
5. Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:2052-2066.e3.

Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. The mean arterial pressure (MAP) takes into account both SBP and DBP readings.

Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
    - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
    - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

References

1. Shibao C, Lipsitz LA, Biaggioni I, American Society of Hypertension Writing Group. Evaluation and treatment of orthostatic hypotension. Journal of the American Society of Hypertension 2013 Jul-Aug;7(4):317-324. DOI: 10.1016/j.jash.2013.04.006.
2. Dalal AS, Van Hare GF. Syncope. In: Kliegman RM, et al., editors. Nelson Textbook of Pediatrics. 22nd ed. Elsevier; 2025:592-599.
3. Fang JC, O'Gara PT. History and physical examination: an evidence-based approach. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:123-140.

Footnotes

A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension, which relies upon blood pressure alone.(1)(2)(3)
B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3):
    - Housing insecurity, as indicated by **1 or more** of the following:
        - Individual or caregiver's current living situation is **1 or more** of the following(4):
            - Does not have own housing (eg, staying in a hotel, shelter, or with others)
            - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
            - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
        - Current housing has **1 or more** of the following:
            - Electrical appliances (eg, stove, refrigerator) not working or unavailable
            - Insufficient heating or cooling
            - Insufficient ventilation
            - Lead paint or pipes
            - Mold
            - Pests (eg, bugs) or rodents
            - Smoke detectors not working or unavailable
    - Food insecurity, as indicated by **1 or more** of the following(5):
        - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
        - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
    - Insufficient transportation, as indicated by **1 or more** of the following(6):
        - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
        - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
    - Insufficient utilities, as indicated by **1 or more** of the following(7):
        - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
        - In the past year, electric, water, gas, or oil company threatened to shut off services.
    - Personal safety risk, as indicated by **2 or more** of the following(5):
        - Individual is sometimes or frequently physically hurt by another person (including family member).
        - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
        - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
        - Individual is sometimes or frequently screamed or cursed at by another person (including family member).

- Insufficient dependent care, as indicated by **1 or more** of the following:
  - In the past year, individual or caregiver was unable to work due to lack of dependent care.
  - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
  - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
  - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
- Depression risk, as indicated by **ALL** of the following(8):
  - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
  - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

References

1. Scanlon A, Reinisch C. Social determinants of health. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:18-20.
2. Billioux A, Verlander K, Anthony S, Alley D. Standardized Screening for Health-Related Social Needs in Clinical Settings: the Accountable Health Communities Screening Tool. [Internet] National Academy of Sciences. 2017 May Accessed at: https://nam.edu/. [accessed 2024 Oct 01]
3. Addressing social determinants of health. In: Perez R, editor. CMSA's Integrated Case Management. 2nd ed. Springer Publishing Company LLC; 2023:62-75.
4. Sandel M, et al. Unstable housing and caregiver and child health in renter families. Pediatrics 2018;14(2):e20172199. DOI: 10.1542/peds.2017-2199.
5. Children's HealthWatch Survey. Screening Instrument [Internet] Children's HealthWatch. 2020 Sep Accessed at: https://childrenshealthwatch.org/. [accessed 2024 Sep 23]
6. PRAPARE®: Protocol for Responding to and Assessing Patient Assets, Risks, and Experiences Screening Tool. [Internet] Association of Asian Pacific Community Health Organizations (AAPCHO) and National Association of Community Health Centers (NACHC). 2016 Sep Accessed at: https://prapare.org/the-prapare-screening-tool/. [accessed 2024Oct 01]
7. Cook JT, et al. A brief indicator of household energy security: associations with food security, child health, and child development in US infants and toddlers. Pediatrics 2008;122(4):e867-75. DOI: 10.1542/peds.2008-0286.
8. Kroenke K, Spitzer RL, Williams JB. The Patient Health Questionnaire-2: validity of a two-item depression screener. Medical Care 2003;41(11):1284-92. DOI: 10.1097/01.MLR.0000093487.78664.3C.

Tachycardia absent

- Tachycardia[A][B] absent, as indicated by **1 or more** of the following(1)(2):
  - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
  - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
  - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
  - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
  - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
  - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Cooney KA, editors. Goldman-Cecil Medicine. 27th ed. Elsevier; 2024:32-35.
2. Cico SJ, Caglar D. Care of the pediatric patient. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:1996-2004.e1.

Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Codes

ICD-10 Diagnosis: C41.2, C79.40, C79.49, D16.6, D32.1, D33.4, D42.1, D43.4, G06.1, G80.0, G80.1, G80.2, G83.4, K59.2, M08.1, M43.05, M43.06, M43.07, M43.15, M43.16, M43.17, M45.5, M45.6, M45.7, M46.45, M46.46, M46.47, M47.15, M47.16, M47.25, M47.26, M47.27, M47.815, M47.816, M47.817, M47.895, M47.896, M47.897, M48.05, M48.061, M48.062, M48.07, M48.8X5, M48.8X6, M48.8X7, M51.05, M51.06, M51.15, M51.16, M51.17, M51.25, M51.26, M51.27, M51.35, M51.85, M54.15, M54.16, M54.17, M96.1, M99.23, M99.33, M99.43, M99.53, M99.63, M99.73, Q76.2, S34.101A, S34.102A, S34.103A, S34.104A, S34.105A, S34.109A, S34.111A, S34.112A, S34.113A, S34.114A, S34.115A, S34.119A, S34.121A, S34.122A, S34.123A, S34.124A, S34.125A, S34.129A, S34.21XA, S34.3XXA [Hide]

ICD-10 Procedure: 00NY0ZZ, 00NY3ZZ, 00NY4ZZ, 01NB0ZZ, 01NB3ZZ, 01NB4ZZ, 01NN0ZZ, 01NN3ZZ, 01NN4ZZ

CPT®: 0275T, 63005, 63012, 63017, 63047, 63052, 63053, 63170, 63200, 63252, 63267, 63272, 63277, 63282, 63287, 63290 [Hide]

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